Christian J. Tai Udovicic, Ph.D.

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Summary

Dr. Tai Udovicic is a postdoctoral scholar investigating the evolution of planetary surfaces and their capacity to produce and retain valuable volatile species like water. He uses a combination of remote sensing spectroscopy and computational modeling, complemented by a background in open source software development to produce insights and tools for active and planned space missions. Tai Udovicic advocates for open science as a board member of the nonprofit OpenPlanetary.

Education

2022 **Northern Arizona University**, Flagstaff, AZ, USA
Ph.D. Astronomy and Planetary Science, *With Distinction*
Dissertation: Eau Claire de la Lune: Clarifying the Origin and Distribution of Water on the Moon

2017 **University of Toronto**, Toronto, ON, Canada
Hon. B.Sc. Planetary Science and Physics, *Cum Laude*.
Thesis: Lunar Surface Maturity in Optical Maturity and Rock Abundance

Publications

[1] E. S. Costello, **C. J. Tai Udovicic**, L. Sun, and P. G. Lucey. "The Rapidly Evolving Surface of Asteroid 4 Vesta Revealed by Rayed Craters and Optical Maturity". In: *Icarus* 454 (2026), p. 117075. DOI: 10.1016/j.icarus.2026.117075.

[2] E. S. Costello, R. R. Ghent, A. Romero-Wolf, P. W. Gorham, P. G. Lucey, **C. J. Tai Udovicic**, P. Linton, A. Ludwig, K. McBride, C. Miki, E. Oberla, J. Rolla, and A. Jung. "Cosmic Rays and the Askaryan Effect Reveal Subsurface Structure and Buried Ice on the Moon". In: *Geophysical Research Letters* 52.6 (2025), e2024GL113304. DOI: 10.1029/2024GL113304.

[3] **C. J. Tai Udovicic**, A. Essunfeld, and E. S. Costello. "Craterpy: Impact Crater Data Science in Python". In: *Journal of Open Source Software* 10.113 (2025), p. 8663. DOI: 10.21105/joss.08663.

[4] M. A. Wieczorek, M. A. Wieczorek, A. Alvarez-Candal, M. Angrisani, N. Attree, A. Bhattacharya, D. T. Blewett, A. Broquet, L. Burkhard, G. Cascioli, G. S. Collins, K. A. Cone, F. Crameri, C. A. Denton, L. Elkins-Tanton, W. Fa, J. Filiberto, J. Flahaut, A. Frigeri, C. M. Guimond, Y. G. Hatipoglu, S. A. Jacobson, D. Kim, A. Lucas, J. Maia, L. Mandon, M. Martinot, K. Miljković, L. Montesi, A. Nawal, R. Oran, L. Pan, R. S. Park, N. E. Putzig, S. Rodriguez, M. Roos-Serote, B. C. Root, T. Ruedas, F. Schmidt, J. F. Snape, **C. J. Tai Udovicic**, N. Tosi, W. V. D. Wal, and N. L. Wagner. "Planetary Research -a Diamond Open Access Journal for Planetary Science". In: *ESS Open Archive* (2025). DOI: 10.22541/essoar.174526273.38397330/v1.

[5] M. S. Phillips, **C. J. Tai Udovicic**, J. E. Moersch, U. Basu, and C. W. Hamilton. "HyPyRameter: A Python Toolbox to Calculate Spectral Parameters from Hyperspectral Reflectance Data". In: *The Planetary Science Journal* 5.11 (2024), p. 258. DOI: 10.3847/PSJ/ad81f8.

[6] **C. J. Tai Udovicic**, K. R. Frizzell, G. R. L. Kodikara, M. Kopp, K. M. Luchsinger, A. Madera, M. L. Meier, T. G. Paladino, R. V. Patterson, F. B. Wroblewski, and D. A. Kring. "Buried Ice Deposits in Lunar Polar Cold Traps Were Disrupted by Ballistic Sedimentation". In: *Journal of Geophysical Research: Planets* 128.5 (2023), e2022JE007567. DOI: 10.1029/2022JE007567.

[7] W. H. Farrand, C. S. Edwards, and **C. J. Tai Udovicic**. "Spectral Evidence for a Pyroclastic Mantle over the Tacquet Formation and Menelaus Domes of Southwest Mare Serenitatis". In: *Icarus* 382 (2022), p. 115021. DOI: 10.1016/j.icarus.2022.115021.

[8] A. P. Jordan, M. L. Shusterman, and **C. J. Tai Udovicic**. "Modeling the Production of Submicroscopic Iron in the Lunar Highlands". In: *Icarus* 387 (2022), p. 115184. DOI: 10/grwr42.

[9] B. E. McKeeby, M. S. Ramsey, **C. J. Tai Udovicic**, C. Haberle, and C. S. Edwards. "Quantifying Sub-Meter Surface Heterogeneity on Mars Using Off-Axis Thermal Emission Imaging System (THEMIS) Data". In: *Earth and Space Science* 9.10 (2022), e2022EA002430. DOI: 10.1029/2022EA002430.

[10] S. Diniega, J. Castillo-Rogez, I. Daubar, J. Filiberto, T. Goudge, K. Lynch, A. Rutledge, J. Rathbun, J. Scully, R. Smith, C. Richey, **C. Tai Udovicic**, and M. Villarreal. "Ensuring a Safe and Equitable Workspace: The Importance and Feasibility of a Code of Conduct, along with Clear Policies Regarding Authorship and Team Membership". In: *Bulletin of the AAS* 53.4 (2021). DOI: 10.3847/25c2cfeb.414c64ae.

[11] **C. J. Tai Udovicic**, E. S. Costello, R. R. Ghent, and C. S. Edwards. "New Constraints on the Lunar Optical Space Weathering Rate". In: *Geophysical Research Letters* 48.14 (2021). DOI: 10.1029/2020GL092198.

[12] R. R. Ghent, L. M. Carter, J. L. Bandfield, **C. J. Tai Udovicic**, and B. A. Campbell. "Lunar Crater Ejecta: Physical Properties Revealed by Radar and Thermal Infrared Observations". In: *Icarus* 273 (2016), pp. 182–195. DOI: 10/f8n4jm.

Selected Conferences

Tai Udovicic, C. J., Costello, E. S., Ghent, R. R., et al. (2025). Unveiling Buried Ice and Subsurface Structure on the Moon with the Cosmic Ray Lunar Sounder (CoRaLS), LPSC, #2860

Tai Udovicic, C. J., Costello, E. S. (2024). Vanishing Rays on Vesta - New Insights into the Impact Gardening Rate using Fresh Craters, AGU, #P41F-2970

Tai Udovicic, C. J., Costello, E. S. (2024). Modernizing Craterpy: An Open-Source Tool for Studying the Moon, Vesta, and Beyond, ELS

Tai Udovicic, C. J., Edwards, C. S., Ghent, R. R. (2023). Submicroscopic Iron Accumulation is Correlated with Local Iron Content, 54th LPSC, #2832

Tai Udovicic, C. J., Gonzales, J. K., Ruiz, J. A., Edwards, C. S., Haberle, C., Farrand, W. H., Bandfield, J. L. (2021). Predicting Emitted Radiance of Rough Lunar Surfaces to Inform Volatile Distribution, GSA, #108-7

Tai Udovicic, C. J., Costello, E. S., Edwards, C. S., Ghent, R. R. (2020). The lunar space weathering rate: dichotomies in agents and products, AGU FM, #P054-0007

Tai Udovicic, C. J., Bandfield, J. L., et al. (2020). Testing Lunar Near-Infrared Corrections with Diviner Observations, 51st LPSC, #2632

Tai Udovicic, C. J., Edwards, C. S., Bandfield, J. L. (2019). Validating a Lunar Roughness-Based Thermal Correction with Diviner Temperature Observations, AGU FM, #P33F-3500

Selected Grants and Awards

Fall 2025	**RCUH outstanding Employee of the Year**, University of Hawai'i
2023–pres.	**Co-I, L-MAPS, NASA Lunar Terrain Vehicle**, University of Hawai'i
2023–pres.	**Co-I, NASA Lunar Data Analysis Program**, University of Hawai'i
Fall 2024	**PI, Catalyst Award for Science Advancement**, University of Hawai'i
2021–2023	**FI, NASA FINESST Research Grant**, Northern Arizona University
Summer 2021	**Exploration Science Summer Internship**, Lunar and Planetary Institute
Fall 2020	**Richard Hayes Memorial Scholar**, Northern Arizona University
Spring 2017	**UC Merit Award**, University of Toronto
Summer 2016	**NSERC Undergraduate Student Research Award**, University of Toronto
Summer 2015	**LPI Summer Internship**, Lunar and Planetary Institute

Teaching

Fall 2024	**SOEST, University of Hawai'i at Manoa** Instructor, *SOEST Open Science and Data Management Plan Workshop* (`https://doi.org/10.5281/zenodo.14188968`)
2018–2021	**Department of Astronomy and Planetary Science, NAU** Instructor, *SPIRL: Scientific Programming In Real Life* (`https://cjtu.github.io/spirl`)
Spring 2021	**Department of Astronomy and Planetary Science, NAU** Instructor, *AST181: Introduction to Observational Astronomy (Lab)*
2016–2017	**Solar System Exploration Group, University of Toronto** Instructor, *Python for Planetary Science, Git for Reproducible Research*

Professional Activities

Board Member	OpenPlanetary (2023–pres.)
Science Teams	CSA: VMMO, LRM, NASA: L-MAPS, LRO Diviner, Lunar Trailblazer
Peer Reviewer	AAS Planetary Science Journal (PSJ), AGU Earth and Space Science (ESS), AGU Geophysical Research Letters (GRL), NASA ROSES proposals (x3)
Organizer	Planetary Data Workshop (2021, 2023), LSSW 13 (2022)
Mentorship	Research Experience for Undergraduates, Undergraduate Research Assistants, Louis Stokes Alliance for Minority Participation